

BLUESTOCK MUTUAL FUND ANALYTICS REPORT

1. Executive Summary

This project analyzes mutual fund performance, investor behavior, and market trends using Python, SQL, and data visualization tools. The goal is to evaluate fund performance, risk, and investor patterns using real financial metrics.

2. Data Sources

- NAV History Data (2022–2026)
- Investor Transactions Data
- SIP Inflows Data
- Portfolio Holdings Data
- Fund Performance Dataset

3. ETL Pipeline

Data was extracted from CSV files, cleaned using Python (Pandas), and loaded into SQLite database. Fact tables include NAV, transactions, and performance data.

4. Exploratory Data Analysis (EDA)

- SIP inflows showed continuous growth over time.
- AUM is dominated by SBI Mutual Fund.
- Small-cap funds showed higher volatility.
- Investor participation increased steadily from 2022–2025.

5. Performance Analysis

Key metrics used:

- CAGR (Compound Annual Growth Rate)
- Sharpe Ratio (Risk-adjusted returns)
- Alpha & Beta (Market comparison)

Findings:

- Equity funds delivered higher returns than debt funds.
- Direct plans outperform regular plans.

6. Risk Analysis

- VaR (95%) calculated to estimate maximum expected loss.
- CVaR used to measure extreme downside risk.
- High volatility observed in small-cap funds.

7. Dashboard Summary

Interactive dashboards were created for:

- SIP trend analysis
- AUM growth comparison
- Investor demographics
- Category inflow heatmap

8. Key Insights

- SBI Mutual Fund leads in AUM.
- SIP inflows reached record highs in 2025.
- Mid-cap funds offer balanced risk-return.
- Investor participation is increasing in T30 cities.

9. Limitations

- Static dataset (no real-time API)
- External market factors not included
- Limited historical depth for some schemes

10. Recommendations

- Increase SIP consistency among investors
- Diversify across categories
- Focus on low drawdown funds for risk-averse investors